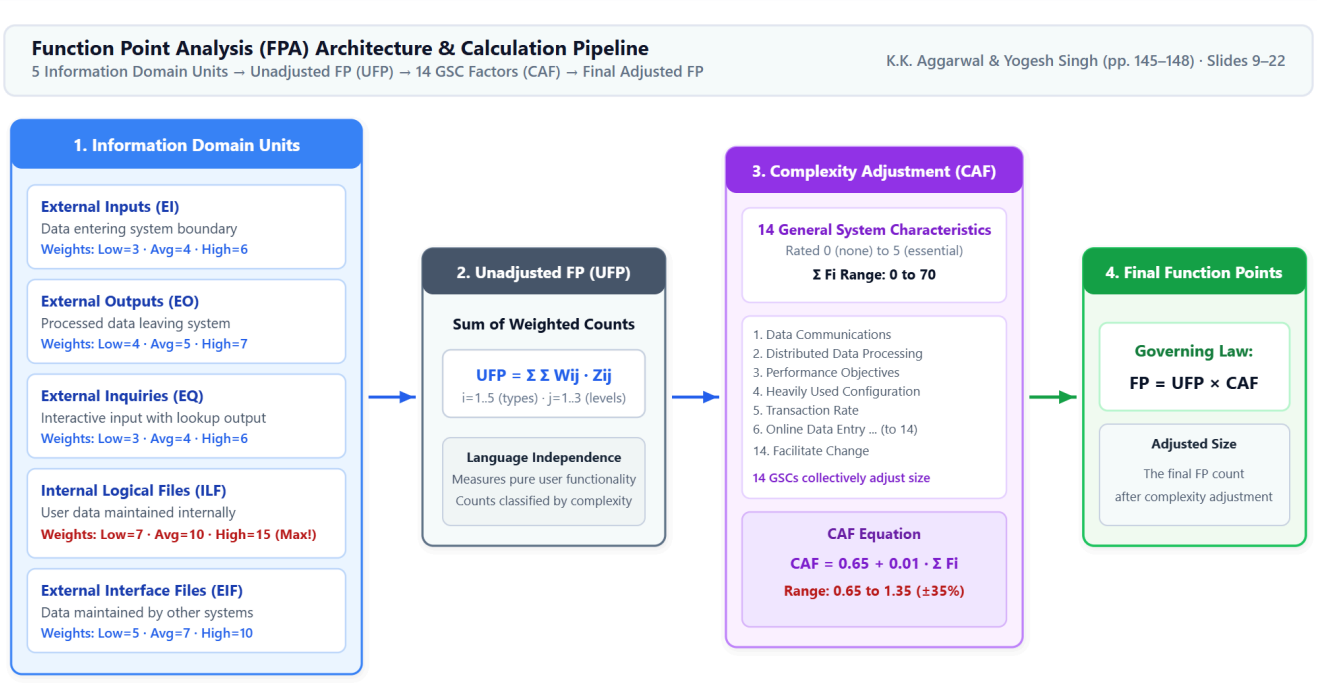
ASE CHAPTER 4

T4 Architectural Diagrams & Visual Models Pack · Software Project Planning

Section 1: Core Architectural & Mathematical Visual Units (5 Units)

Diagram 01. Function Point Analysis (FPA) Architecture & Calculation Pipeline

Diagram and specify the architectural pipeline of Albrecht's Function Point Analysis (FPA), illustrating how the 5 information domain characteristics, Unadjusted Function Points (UFP), and 14 General System Characteristics (CAF) synthesize into final adjusted Function Points.



Function Point Analysis Architecture

Architectural & Mathematical Walkthrough:

- **1. Step 1 — Information Domain Units (5 Functional Characteristics):** The system's functional boundary is decomposed into five components:
- **External Inputs (EI):** Transactions capturing external data that alters internal files (Weights: Low=3, Average=4, High=6).
- **External Outputs (EO):** Transactions generating processed reports or derived calculations leaving the system (Weights: Low=4, Average=5, High=7).
- **External Inquiries (EQ):** Interactive input-output lookups that retrieve data without altering internal state (Weights: Low=3, Average=4, High=6).

- **Internal Logical Files (ILF):** User-identifiable data groups maintained entirely inside the application boundary (Weights: Low=7, Average=10, High=15 — the highest weight in FPA).
- **External Interface Files (EIF):** Reference data files maintained by external systems across interfaces (Weights: Low=5, Average=7, High=10).
- **2. Step 2 — Unadjusted Function Points (UFP) Calculation:** Each identified component is multiplied by its complexity weight and summed across all five categories:

$$\text{UFP} = \sum_{i=1}^{5} \sum_{j=1}^{3} W_{ij} \cdot Z_{ij}$$

- **3. Step 3 — Complexity Adjustment Factor (CAF) Scaling:** The system is rated against **14 General System Characteristics (GSCs)** (F_1 to F_{14}) on a 0 to 5 scale ($\sum F_i$ ranges from 0 to 70). The Complexity Adjustment Factor is computed as:

$$\text{CAF} = 0.65 + 0.01 \times \sum_{i=1}^{14} F_i$$

  This linearly scales the baseline UFP within a bounded range of 0.65 **to** 1.35 **(**$\pm 35\%$**)**.
- **4. Step 4 — Final Function Points & Metrics:** The final adjusted size is computed as $\text{FP} = \text{UFP} \times \text{CAF}$. The output is the adjusted function-point size estimate. Any later use in productivity or quality measures requires separately defined project data

Source: K.K. Aggarwal & Yogesh Singh (3rd ed.), §4.3.2, pp. 145–148; Companion Slides 9–22.

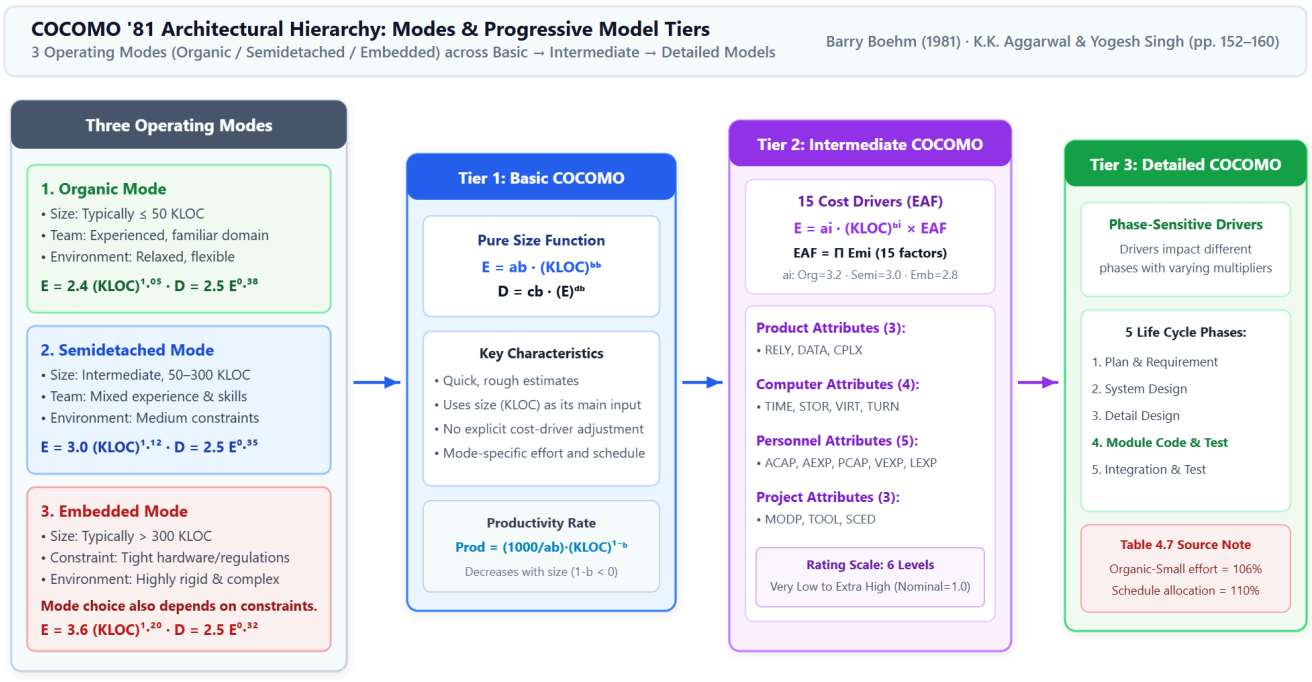
ملاحظة أكاديمية هامة (Academic Note)

ملاحظة الطالب والفخ الامتحاني:

- في الامتحان، احرص على تذكر أوزان المستوى المتوسط (Average): المدخلات 4، المخرجات 5، الاستعلامات 4، الملفات الداخلية 10، ملفات الواجهات 7.
- معامل التعديل (CAF) لا يمكن أن يقل عن 0.65 ولا أن يزيد عن 1.35؛ فإذا حصلت على رقم خارج هذا النطاق فاعلم أن حساباتك خاطئة.

Diagram 02. COCOMO '81 Architectural Hierarchy: Modes & Model Tiers

Diagram and explain the structural hierarchy of Barry Boehm's COCOMO '81 model, showing the progression across the three operating modes and three model tiers.



COCOMO '81 Hierarchy

Architectural Breakdown:

- **1. The Three Operating Modes (Environmental Complexity):**
- **Organic Mode:** Typically smaller projects with experienced teams, familiar domains, and relatively flexible constraints.

$$\text{Effort: } E = 2.4(\text{KLOC})^{1.05}\ \text{PM}, \quad \text{Schedule: } D = 2.5(E)^{0.38}\ \text{Months}$$

- **Semidetached Mode:** Typically intermediate projects with mixed team experience and constraints.

$$\text{Effort: } E = 3.0(\text{KLOC})^{1.12}\ \text{PM}, \quad \text{Schedule: } D = 2.5(E)^{0.35}\ \text{Months}$$

- **Embedded Mode:** Projects operating under tight hardware, software, or operational constraints.

$$\text{Effort: } E = 3.6(\text{KLOC})^{1.20}\ \text{PM}, \quad \text{Schedule: } D = 2.5(E)^{0.32}\ \text{Months}$$

- **2. Tier 1 — Basic COCOMO:** Single-variable power function of size alone. Useful as an initial estimate when only size and mode are available.
- **3. Tier 2 — Intermediate COCOMO:** Modifies nominal effort by the **Effort Adjustment Factor (EAF)** derived from **15 cost drivers** grouped into Product (3), Computer (4), Personnel (5), and Project (3) attributes:

$$E = a_i(\mathrm{KLOC})^{b_i} \times \prod_{j=1}^{15} EM_j \quad (a_{\text{organic}} = 3.2, a_{\text{semi}} = 3.0, a_{\text{embedded}} = 2.8)$$

- **4. Tier 3 — Detailed COCOMO:** Acknowledges that cost drivers impact phases unevenly. Employs **phase-sensitive effort multipliers** to allocate effort by phase. Table 4.7 includes a Plan & Requirement row and reports raw Organic-Small percentages that sum to 106% for effort and 110% for schedule; it should not be presented as a normalized five-phase 100% distribution.

(Table 4.7 source values are preserved as printed: the listed Organic-Small effort entries sum to 106%, and the schedule entries sum to 110%.)

Source: K.K. Aggarwal & Yogesh Singh (3rd ed.), §4.4.1–§4.4.3, pp. 152–160; Companion Slides 41–65.

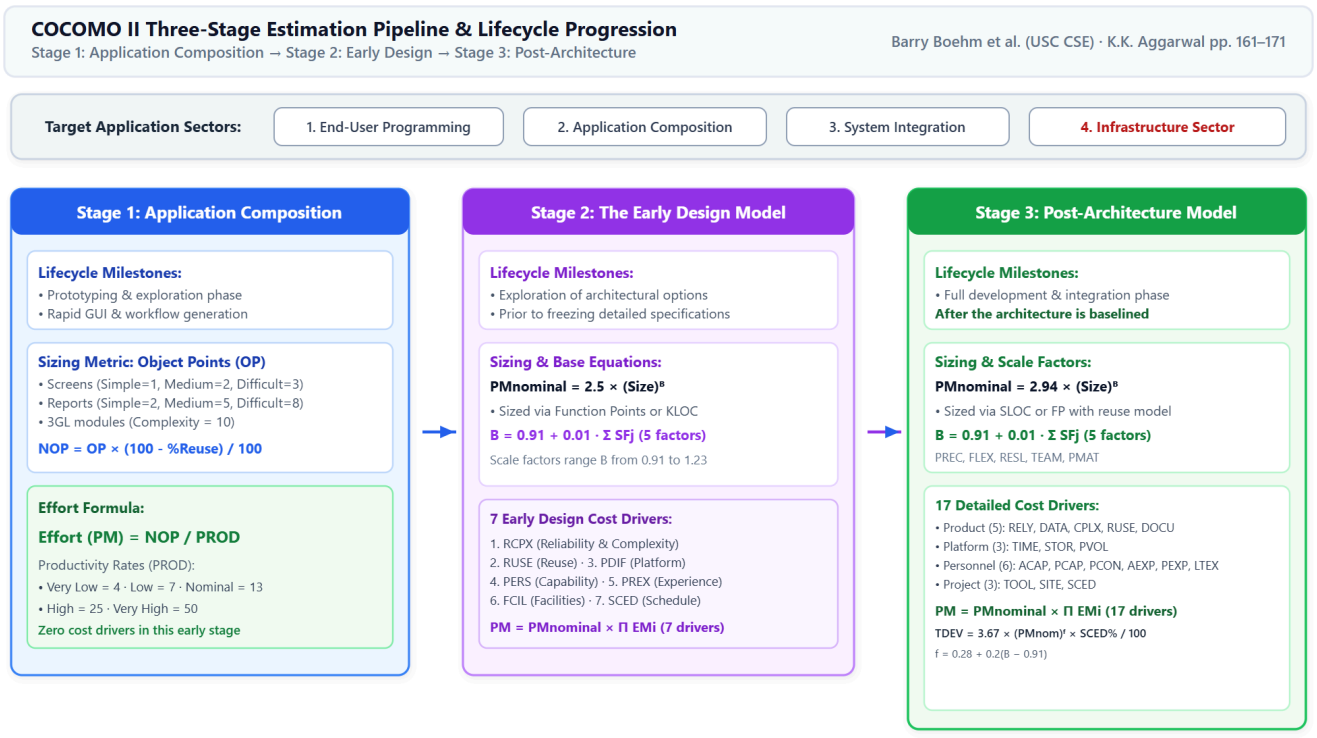
ملاحظة أكاديمية هامة (Academic Note)

ملاحظة الطالب والفخ الامتحاني:

- لاحظ أن معامل الأساس (a_i) في Intermediate COCOMO يختلف عن (a_b) في Basic:
- في Organic: يتحول المعامل من 2.4 إلى 3.2.
- في Semidetached: يبقى 3.0.
- في Embedded: يتحول من 3.6 إلى 2.8.
- الأس (b) يبقى ثابتاً (1.05، 1.12، 1.20) في كلا النموذجين.

Diagram 03. COCOMO II Three-Stage Lifecycle Estimation Pipeline

Diagram the three estimation stages of COCOMO II, illustrating lifecycle milestones, sizing metrics, scale factor mechanics, and cost driver allocations.



COCOMO II Three-Stage Pipeline

Lifecycle Pipeline Breakdown:

- **1. Target Project Categories (Top Banner):** COCOMO II identifies four application categories: End-User Programming, Application Composition, System Integration, and the **Infrastructure Sector** (deep systems software such as OS, DBMS, compilers).
- **2. Stage 1 — Application Composition Model:**
- **When Applied:** Prototyping and UI exploration phase.
- **Sizing: Object Points (OP)** derived from screen, report, and 3GL module counts, adjusted for reuse: $\mathrm{NOP} = \mathrm{OP} \times (100 - \%\mathrm{Reuse})/100$.
- **Effort Equation:** $\mathrm{Effort\ (PM)} = \frac{\mathrm{NOP}}{\mathrm{PROD}}$, where PROD ranges from 4 (Very Low) to 50 (Very High).
- **3. Stage 2 — The Early Design Model:**
- **When Applied:** High-level architectural exploration before detailed subsystem specifications are frozen.
- **Sizing:** Unadjusted Function Points (UFP) converted to KLOC.
- **Base Formula:** $\mathrm{PM}_{\mathrm{nominal}} = 2.5 \times (\mathrm{Size})^{B}$, where exponent B is derived from **5 Scale Factors** (SF_j):

$$B = 0.91 + 0.01 \times \sum_{j=1}^{5} SF_j \quad (B \text{ ranges from } 0.91 \text{ to } 1.23)$$

- **Cost Drivers:** Adjusted using **7 Early Design Effort Multipliers**: RCPX, RUSE, PDIF, PERS, PREX, FCIL, SCED.
- **4. Stage 3 — Post-Architecture Model:**
- **When Applied:** Full development and integration phase after software architecture is baselined.
- **Sizing:** Source Lines of Code (SLOC) or Function Points with comprehensive reuse formulas.
- **Cost Drivers:** Adjusted using **17 detailed Post-Architecture Cost Drivers** spanning Product (5), Platform (3), Personnel (6), and Project (3) attributes.
- **Schedule Formula:** $\text{TDEV} = [3.67 \times (\text{PM}_{\text{nominal}})^{F}] \times \frac{\text{SCED\%}}{100}$, where $F = 0.28 + 0.2(B - 0.91)$.

Source: K.K. Aggarwal & Yogesh Singh (3rd ed.), §4.5, pp. 161–171; Companion Slides 73–86.

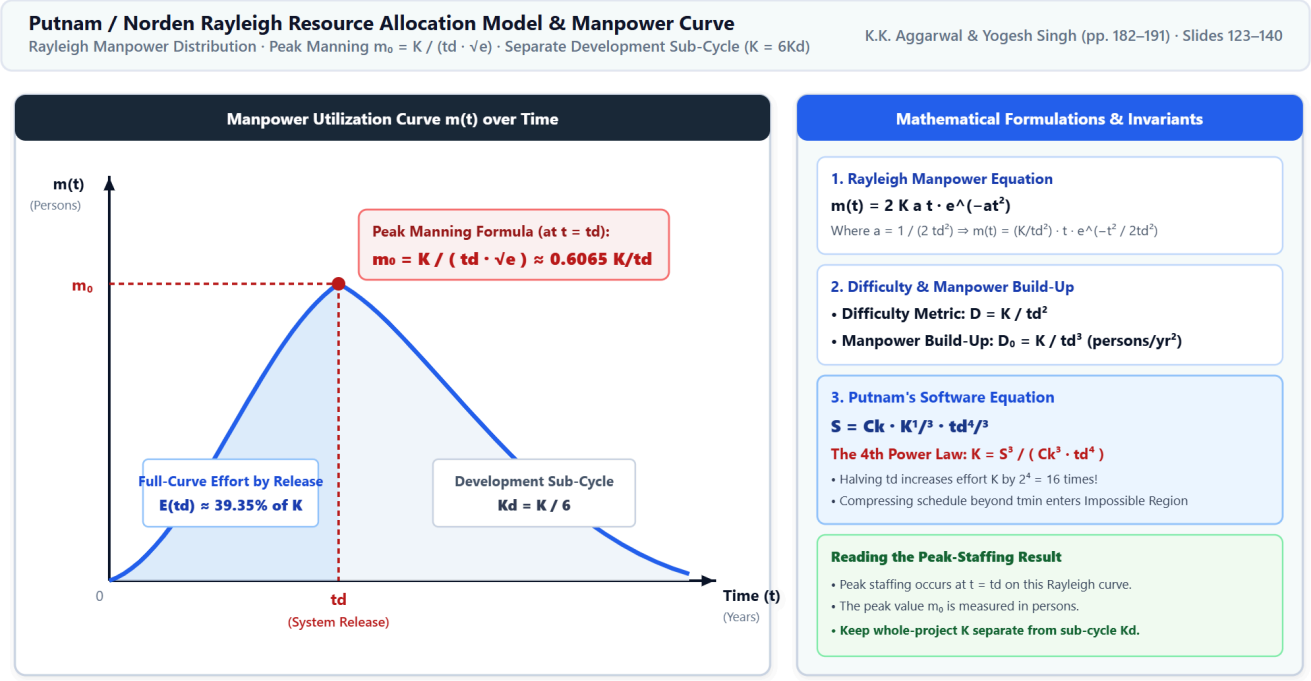
ملاحظة أكاديمية هامة (Academic Note)

ملاحظة الطالب والفخ الامتحاني:

- في COCOMO II، عوامل المقياس الخمسة (PREC, FLEX, RESL, TEAM, PMAT) تُستخدم في كل من مرحلة Early Design ومرحلة Post-Architecture لحساب الأس B.
- الاختلاف بين المرحلتين يكمن في محركات الكلفة: 7 محركات مجمعة في Early Design مقابل 17 محركاً مفصلاً في Post-Architecture.

Diagram 04. Putnam / Norden Rayleigh Resource Allocation Model & Curve

Diagram and mathematically specify the Putnam Resource Allocation Model, illustrating the Rayleigh manpower curve, the peak-manning point m_0, cumulative effort by t_d, the separate development sub-cycle, and the fourth-power schedule law.



Putnam Rayleigh Resource Allocation Model

Mathematical & Visual Walkthrough:

- **1. The Rayleigh Manpower Equation:** Manpower utilization $m(t)$ over time is represented by the Norden/Rayleigh staffing equation:

  $$m(t) = 2Kat \cdot e^{-at^2}$$

  Substituting the shape parameter $a = \frac{1}{2t_d^2}$ yields:

  $$m(t) = \frac{K}{t_d^2} \cdot t \cdot e^{-\frac{t^2}{2t_d^2}}$$

- **2. Peak Manning Verification (At $t = t_d$):** Setting $t = t_d$ yields maximum team staffing:

  $$m_0 = m(t_d) = \frac{K}{t_d^2} \cdot t_d \cdot e^{-1/2} = \frac{\mathbf{K}}{\mathbf{t_d\sqrt{e}}} \approx \mathbf{0.6065}\frac{\mathbf{K}}{\mathbf{t_d}}$$

  Formula Check: The peak is $m_0 = \frac{K}{t_d\sqrt{e}}$. Example 4.12 (printed p. 184) uses $K = 95$ person-years and $t_d = 1.75$ years, giving about 33 persons. Example 4.13 (printed p. 185) uses $K = 600$ person-years and $t_d = 3.5$ years, giving about 104 persons. Both source examples confirm the radical.

- **3. Effort accumulated by release versus development sub-cycle:** Integrating the full-project curve to t_d gives $E(t_d) = K(1 - e^{-1/2}) \approx 0.3935K$. Total development-sub-cycle effort is separately defined by $K = 6K_d$; do not label these two quantities as the same.
- **4. Governing Metrics:**
- **Difficulty Metric:** $D = \frac{K}{t_d^2}$ (persons/year).
- **Manpower Build-Up Metric:** $D_0 = \frac{K}{t_d^3}$ (persons/year^2).
- **Putnam's Software Equation & 4th Power Law:**

$$S = C_k \cdot K^{1/3} \cdot t_d^{4/3} \implies K = \frac{S^3}{C_k^3 \cdot t_d^4} \implies \mathbf{K \propto \frac{1}{t_d^4}}$$

Halving delivery time increases effort by $2^4 = 16$ times!

Source: K.K. Aggarwal & Yogesh Singh (3rd ed.), §4.6, printed pp. 182–191; Companion Slides 123–140, 149–165.

ملاحظة أكاديمية هامة (Academic Note)

ملاحظة الطالب والفخ الامتحاني:

- تأكد دائماً من كتابة الجذر التربيعي لـ e في قانون ذروة التوظيف: $m_0 = \frac{K}{t_d\sqrt{e}}$.
- الصعوبة $D = K/t_d^2$، ومعدل بناء القوى العاملة $D_0 = K/t_d^3$. والجهد التراكمي حتى t_d على منحنى المشروع ($0.3935K$) يختلف عن جهد دورة التطوير $K_d = K/6$.

Diagram 05. Software Risk Management Framework & Workflow

Diagram the complete software risk management architecture, detailing the two primary categories (Risk Assessment and Risk Control), their six sequential sub-activities, and the quantitative Risk Exposure formula.



Software Risk Management Framework

Architectural Breakdown:

- **1. Core Risk Definition:** A risk is a problem that may cause loss or threaten project success but has not happened yet.
- **2. Category 1 — Risk Assessment (Evaluation Phase):**
- **Risk Identification:** Uncovers possible threats using a checklist of common risk areas or an organizational database of previously identified risks.
- **Risk Analysis:** Examines how project outcomes might change when risk input variables change.
- **Risk Prioritization:** Orders risks by computing **Risk Exposure (RE)**:

$$\mathrm{RE} = \mathbf{P} \times \mathbf{L}$$

  Give attention first to the risks with the largest estimated exposure.
- **3. Category 2 — Risk Control (Mitigation & Monitoring Phase):**
- **Risk Management Planning:** Prepares a plan for dealing with each significant risk and records the decision.
- **Risk Resolution:** Executes the plan for dealing with the risk.
- **Risk Monitoring:** Tracks risks during the project.

Source: K.K. Aggarwal & Yogesh Singh (3rd ed.), §4.7.3, pp. 196–198; Companion Slide 190.

> 📝 **ملاحظة أكاديمية هامة (Academic Note)**
>
> **ملاحظة الطالب والفخ الامتحاني:**
>
> - احفظ التقسيم والأنشطة الستة كما تظهر في الشكل؛ الشريحة تعرض هيكل الأنشطة ولا تثبت حلقة تغذية راجعة أو RMMM.

Topic-to-Question Map

Diagram / Unit	Topic Area	Primary Course Sources	Key Visual Invariants & Formulas
Diagram 01	Size Estimation (FPA)	K.K. Aggarwal pp. 145–148; Slides 9–22	5 Domain Units, UFP weighted sum, 14 GSC factors ($\pm 35\%$), $\mathrm{FP} = \mathrm{UFP} \times \mathrm{CAF}$.
Diagram 02	COCOMO '81 Architecture	K.K. Aggarwal pp. 152–160; Slides 41–65	3 Modes (Organic, Semi, Embedded), Basic vs Intermediate (15 drivers), Detailed phase-sensitive multipliers.
Diagram 03	COCOMO II Pipeline	K.K. Aggarwal pp. 161–171; Slides 73–86	4 application sectors, Stage 1 (Object Points/NOP), Stage 2 (Early Design), Stage 3 (Post-Architecture).
Diagram 04	Putnam Resource Model	K.K. Aggarwal printed pp. 182–191; Slides 123–140, 149–165	Norden/Rayleigh Curve, Peak manning $m_0 = \frac{K}{t_d\sqrt{e}}$, full-curve cumulative effort by release, separate development sub-cycle, 4th-power law ($K \propto t_d^{-4}$).
Diagram 05	Software Risk Framework	K.K. Aggarwal pp. 196–198; Slides 190–191	Risk Assessment and Risk Control, their six activities, and Risk Exposure $\mathrm{RE} = P \times L$.
